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TN SKILLS REPORT OCT-2026

AI-Augmented Backend Development with Node.js and MongoDB

Project title:
Date:

Trainer **External Examiner**

TN SKILLS PROJECT

PROJECT TITLE: AI FAQ ASSISTANT API

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1.INTRODUCTION

The AI FAQ Assistant is an intelligent question-answering system designed to provide quick and relevant answers to frequently asked questions. In many organizations and educational institutions, users repeatedly ask similar questions related to services, procedures, facilities, and general information. Answering these questions manually can consume time and effort. The proposed system helps to reduce this workload by providing an automated platform for handling frequently asked questions.

The system allows users to submit their questions through an application interface. The submitted question is processed by the backend system and analyzed to identify the required information. The system uses Artificial Intelligence and Natural Language Processing (NLP) techniques to understand the user's question and generate or retrieve a suitable response. This makes the interaction more convenient because users can ask questions using natural language instead of searching through large amounts of information manually.

The application is developed using Node.js and Express.js for backend processing. MongoDB is used to store FAQ information and user-related data. The system also integrates an AI service to process user queries and provide intelligent responses. The application is organized into different modules such as authentication, FAQ management, AI processing, database configuration, middleware, routes, controllers, and services. This modular structure makes the application easier to develop, maintain, and extend.

The system provides features such as user registration and authentication, FAQ creation and retrieval, database management, AI-based question answering, and API-based communication. The APIs can be tested using tools such as Postman to verify whether the requests and responses are working correctly. When a user submits a question, the system processes the request and returns the appropriate response through the API.

The main objective of the AI FAQ Assistant is to provide a fast, simple, and automated method of answering frequently asked questions. By combining backend technologies, database management, and Artificial Intelligence, the system demonstrates how intelligent assistants can be developed to improve information accessibility and reduce repetitive manual work.

2.PROJECT OVERVIEW

Project Description

The AI FAQ Assistant is an intelligent application developed to provide automated answers to frequently asked questions. It helps users get quick and relevant information without depending on manual assistance. The system uses Artificial Intelligence to understand user queries and provide suitable responses.

Technologies Used

The application is developed using Node.js and Express.js for backend processing and API management. MongoDB is used to store user details and FAQ information. An AI service is integrated to process user questions and generate appropriate answers. Postman is used to test and verify the API requests and responses.

Working Process

The user submits a question through the system. The request is received by the appropriate API route and processed by the backend. The system retrieves relevant FAQ information from the database or sends the question to the AI service for processing. The resulting answer is then returned to the user.

Key Features

The major features include user registration and authentication, FAQ creation and retrieval, AI-based question answering, MongoDB database integration, REST API handling, and error management. The project provides a simple and efficient way to automate frequently asked questions and demonstrates the practical use of AI in modern application development.

System Modules

The project consists of several important modules. The User Authentication Module manages user registration and authentication. The FAQ Management Module allows FAQ questions and answers to be created, stored, and retrieved from the database.

3.SYSTEM ARCHITECTURE

The AI FAQ Assistant follows a modular client-server architecture. It consists of different components that work together to receive user requests, process the information, store data, and provide suitable answers.

User Layer

The user interacts with the system by registering, logging in, submitting FAQ-related requests, and asking questions. These requests are sent to the backend through APIs.

API Layer

The Node.js and Express.js backend receives and manages requests from the user. Separate API routes are used for authentication, FAQ management, and AI-based question answering.

Controller Layer

The controller layer contains the main application logic. The authentication controller handles registration and login, the FAQ controller manages FAQ operations, and the AI controller processes questions submitted for AI-based answers.

Service and Database Layer

The service layer communicates with the AI service to process user questions and generate suitable responses. MongoDB is used to store user details and FAQ information. The database configuration module manages the connection between the application and MongoDB.

Architecture Flow

User → API Routes → Controllers → Services / MongoDB → AI Processing → Response → User

This architecture makes the AI FAQ Assistant organized, modular, maintainable, and suitable for handling authentication, FAQ management, and AI-based question answering.

4.SETUP INSTRUCTIONS

Software Installation

Install Node.js, MongoDB, and Visual Studio Code on the system. Node.js is required to run the backend application, while MongoDB is used for storing project data.

Project Setup

Open the project folder in Visual Studio Code and open the terminal. Run `npm install` to install all the required dependencies mentioned in the package.json file.

Database Configuration

Start the MongoDB service and configure the database connection in the `.env` file. The application uses MongoDB to store user details and FAQ information.

AI Configuration

Add the required AI service API key in the `.env` file. This allows the system to process user questions and generate AI-based answers.

Run the Project

Start the application using the required npm command. The Express.js server runs on port 5000. Postman can be used to test the authentication, FAQ, and AI APIs and verify their responses.

API Testing

Use Postman to test the authentication, FAQ, and AI APIs. Send requests to the required endpoints and verify the responses returned by the server.

4.FOLDER STRUCTURE

```
│
├── node_modules
│
├── src
│   ├── config
│   │   └── db.js
│   │
│   ├── controllers
│   │   ├── aiController.js
│   │   ├── authController.js
│   │   └── faqController.js
│   │
│   ├── middleware
│   │   ├── authMiddleware.js
│   │   ├── errorMiddleware.js
│   │   └── validationMiddleware.js
│   │
│   ├── models
│   │
│   ├── routes
│   │   ├── aiRoutes.js
│   │   ├── authRoutes.js
│   │   └── faqRoutes.js
│   │
│   ├── services
│   │   └── geminiService.js
│   │
│   ├── utils
│   │   └── helpers.js
│   │
│   ├── app.js
```

```
│   └── server.js
│
├── .env
├── AI_FAQ_Assistant_API.postman_collection...
├── package-lock.json
├── package.json
├── README.md
└── test_api.js
```

5.RUN THE APPLICATION

Start MongoDB

Before running the application, make sure that the MongoDB service is running. MongoDB is required because the application stores user details and FAQ information in the database.

Open the Project

Open the AI FAQ Assistant project folder in Visual Studio Code. Open the integrated terminal and make sure the terminal is opened in the project directory.

Install Dependencies

Install all the required Node.js packages by running the `npm install` command in the terminal. This installs the dependencies required for the Express.js application.

Start the Server

After completing the setup, start the application using the appropriate npm start command. The Node.js and Express.js server starts running on the configured port, such as port 5000.

Test the APIs

Open Postman and enter the required API endpoint. Select the appropriate HTTP method such as POST or GET and provide the required input data. Send the request and check the response returned by the server.

Verify the Output

The application processes the request and returns the appropriate response. User authentication, FAQ management, and AI-based question answering can be verified through the API responses.

Application Execution

Once the server, database, and AI service are properly connected, the AI FAQ Assistant is ready to handle user requests. The application can receive questions, process them, access FAQ data, and generate suitable AI-based responses.

6.OBJECTIVE

The main objective of the AI FAQ Assistant is to develop an intelligent system that can provide quick and suitable answers to frequently asked questions.

Automated Question Answering

To provide automated answers to user questions using Artificial Intelligence and reduce the need for manual assistance.

FAQ Management

To create, store, and retrieve frequently asked questions and their answers using a MongoDB database.

User Authentication

To provide secure user registration and login functionality for accessing the application.

AI Integration

To integrate an AI service that can process user queries and generate relevant responses.

API-Based Communication

To develop REST APIs using Node.js and Express.js for handling user authentication, FAQ operations, and AI-based question answering.

Efficient Information Access

To help users obtain required information quickly through a simple and efficient question-answering system.

Overall, the project aims to demonstrate how Artificial Intelligence, backend technologies, and database management can be combined to develop an automated FAQ assistance system.

7.SCOPE OF THE PROJECT

The AI FAQ Assistant is designed to provide automated and intelligent answers to frequently asked questions. The system combines backend technologies, database management, and Artificial Intelligence to provide an efficient question-answering solution.

Automated FAQ Assistance

The system can handle frequently asked questions and provide suitable answers without requiring continuous manual support.

User Management

The project supports user registration and authentication, allowing users to access the available services securely.

FAQ Management

Users can create and retrieve FAQ information. The FAQ data is stored and managed using MongoDB for efficient access.

AI-Based Responses

The system can process questions that are not directly available in the stored FAQ information by using an integrated AI service to generate suitable responses.

API-Based Application

The project provides APIs for authentication, FAQ operations, and AI question answering. These APIs can be tested using tools such as Postman.

Future Expansion

The system can be extended with a user-friendly frontend, voice-based interaction, multilingual support, improved AI models, and additional knowledge sources.

Overall, the scope of the project covers automated FAQ management and AI-based question answering while providing a foundation for further development.

8.HARDWARE REQUIREMENTS

The AI FAQ Assistant requires basic computer hardware to develop, run, and test the application.

Processor

A minimum dual-core processor is sufficient to run the Node.js application, MongoDB, and development tools.

RAM

A minimum of 4 GB RAM is recommended for smooth execution. 8 GB RAM provides better performance when running multiple development tools simultaneously.

Storage

At least 10 GB of free storage is recommended for the project files, Node.js packages, MongoDB data, and development software.

Display

A standard monitor with a minimum resolution of 1366 × 768 is sufficient for using Visual Studio Code, MongoDB tools, and Postman.

Internet Connection

A stable internet connection is required for downloading dependencies, accessing the AI service, and testi

9.SOFTWARE REQUIREMENTS

The AI FAQ Assistant requires suitable software tools and technologies for development, database management, AI integration, and API testing.

Operating System

Windows, Linux, or macOS can be used to develop and run the application.

Programming Technologies

Node.js and Express.js are used for backend development and API handling.

Database

MongoDB is used to store and manage user details and FAQ information.

AI Service

An AI service such as Gemini is integrated to process user questions and generate suitable responses.

Development Tool

Visual Studio Code is used to create, edit, and manage the project files.

API Testing Tool

Postman is used to send API requests and verify the responses from the application.

Supporting Tools

npm is used to install and manage Node.js packages. A web browser is used to access and test the application and its API res

10.MODULES DESCRIPTION

The AI FAQ Assistant is divided into different modules, where each module performs a specific function. These modules work together to provide authentication, FAQ management, database operations, and AI-based question answering.

User Authentication Module

This module manages user registration and login. It verifies user details and provides authenticated access to the application.

FAQ Management Module

This module handles the creation and retrieval of frequently asked questions. FAQ information is stored in MongoDB and can be accessed through the appropriate APIs.

AI Question Answering Module

This is the main intelligent module of the system. It receives the user's question and sends it to the integrated AI service. The generated answer is then returned to the user.

Database Module

This module manages the connection between the application and MongoDB. It stores and retrieves user details and FAQ information whenever required.

API Module

The API module provides endpoints for authentication, FAQ operations, and AI question answering. It allows different requests to be processed by the backend.

Error Handling Module

This module handles invalid requests, authentication errors, database errors, and AI service errors. It helps the application return proper error responses and improves system reliability.

11.DATABASE DESIGN

The AI FAQ Assistant uses MongoDB as its database to store and manage the information required by the application. MongoDB is a NoSQL database that stores data in the form of documents and collections.

User Collection

The User collection stores user-related information such as name, email, password, and other required authentication details. This information is used during registration and login.

FAQ Collection

The FAQ collection stores frequently asked questions and their corresponding answers. It may also contain category or other relevant information required for managing FAQs.

Database Connection

The application connects to MongoDB through the database configuration module. The database connection details are maintained in the environment configuration file.

Data Management

The application uses backend APIs to create, retrieve, and manage database records. MongoDB allows the system to efficiently store and access user and FAQ information when required.

Database Flow

User Request → Express.js API → Controller → MongoDB → Response

The database design provides organized data storage and allows the AI FAQ Assistant to access the req

12.API DESIGN

The AI FAQ Assistant uses REST APIs to allow communication between the user requests and the backend application. The APIs are developed using Node.js and Express.js and are tested using Postman.

Authentication API

The authentication API handles user registration and login. It receives user details, validates the information, and returns the appropriate response.

FAQ API

The FAQ API is used to create and retrieve frequently asked questions. It communicates with MongoDB to store and access FAQ information.

AI API

The AI API receives questions from the user and sends them to the integrated AI service for processing. The generated answer is returned through the API response.

API Request and Response

The client sends a request to the appropriate API endpoint using methods such as GET or POST. The server processes the request and returns a response containing the requested information or result.

API Testing

Postman is used to test the available APIs. Different requests are sent with required input data, and the returned responses are verified to ensure that the APIs work correctly.

API Flow

User Request → API

13.IMPLEMENTATION

The AI FAQ Assistant is implemented using Node.js, Express.js, MongoDB, and an integrated AI service. The application is developed using a modular structure, where each component performs a specific function.

Backend Implementation

Node.js and Express.js are used to develop the backend of the application. Express.js manages the API routes and handles requests received from users.

User Authentication

The authentication module is implemented to manage user registration and login. User details are validated and stored in MongoDB, allowing the system to authenticate users when they log in.

FAQ Implementation

The FAQ module provides APIs for creating and retrieving frequently asked questions. The FAQ data is stored in MongoDB and accessed through the backend controllers.

AI Integration

The AI module connects the application with the integrated AI service. When a user submits a question, the request is processed and sent to the AI service. The generated response is then returned to the user through the API.

API Implementation

REST APIs are implemented for authentication, FAQ management, and AI question answering. Postman is used to send requests and verify the responses during development and testing.

Error Handling

Error-handling mechanisms are implemented to manage invalid requests, authentication errors, database connection problems, and AI service errors. This helps the application provide proper responses and maintain

14.TESTING

Testing is performed to verify that the AI FAQ Assistant works correctly and produces the expected results. Each major function of the application is tested using different API requests through Postman.

Authentication Testing

The registration and login APIs are tested with valid and invalid user details. The system is checked to ensure that valid users are authenticated and incorrect details are handled properly.

FAQ Testing

The FAQ APIs are tested by creating and retrieving FAQ information. The responses are verified to ensure that the data is correctly stored in and retrieved from MongoDB.

AI Response Testing

The AI API is tested by submitting different questions. The returned responses are checked to verify that the AI service processes the questions and provides suitable answers.

API Testing

All major API endpoints are tested using Postman. Request methods, input data, status responses, and returned results are verified during testing.

Error Testing

Invalid requests, missing data, database errors, and AI service errors are tested to ensure that the system returns appropriate error messages.

The testing process helps verify the functionality, reliability, and proper integration of the different components of the AI FAQ Assistant.

15.RESULTS AND OUTPUT

The AI FAQ Assistant was successfully implemented and tested using the developed backend APIs. The system performs user authentication, FAQ management, database operations, and AI-based question answering as expected.

User Registration and Login

The registration API successfully accepts user details and stores the required information in MongoDB. The login API verifies the provided credentials and returns the appropriate response for authenticated users.

FAQ Management Output

The FAQ module successfully creates and retrieves frequently asked questions. The stored FAQ information can be accessed through the corresponding API requests.

AI Answer Output

The AI module successfully receives user questions and communicates with the integrated AI service. The generated answer is returned to the user through the API response.

API Execution

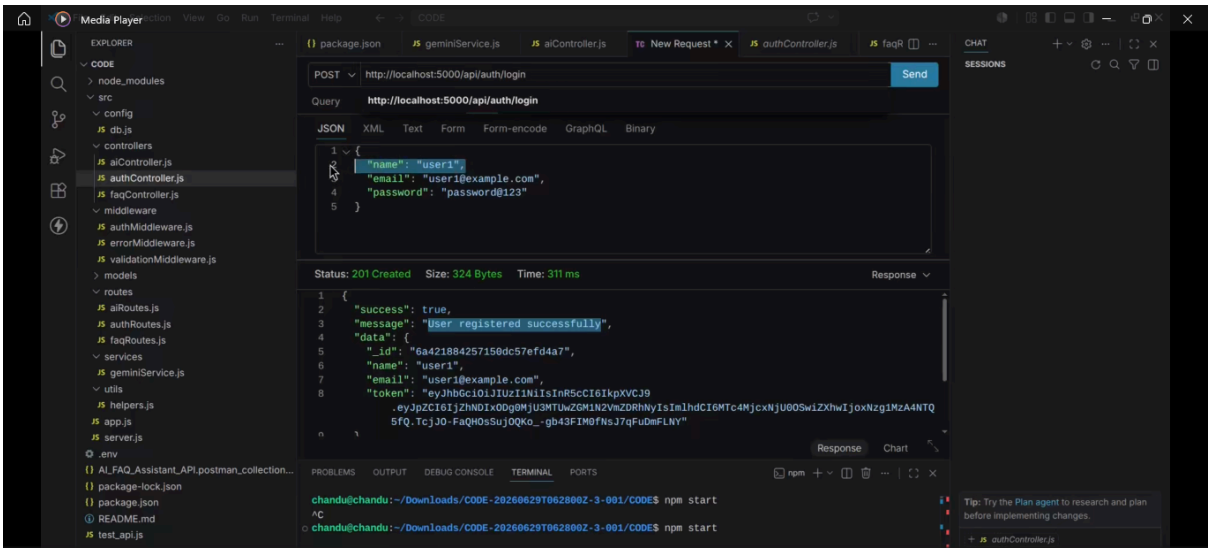
The APIs were tested using Postman by sending different requests and checking the returned responses. The testing confirmed that the major functions of the application work correctly.

Final Result

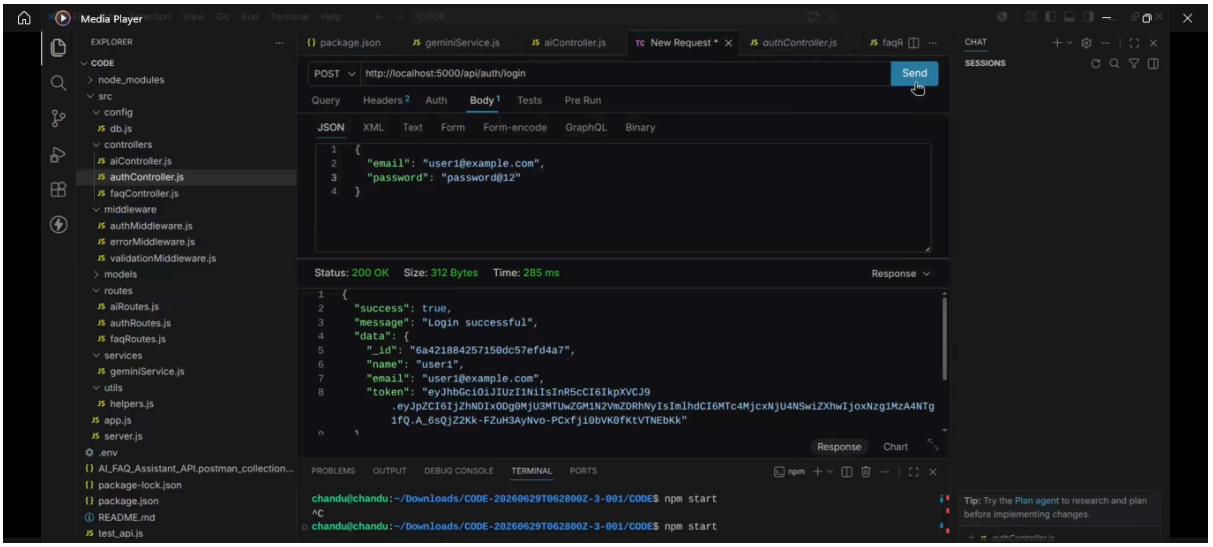
The completed AI FAQ Assistant provides an automated way to manage FAQs and generate AI-based answers. The integration of Node.js, Express.js, MongoDB, and AI services demonstrates the successful implementation of the project.

Output Screenshots

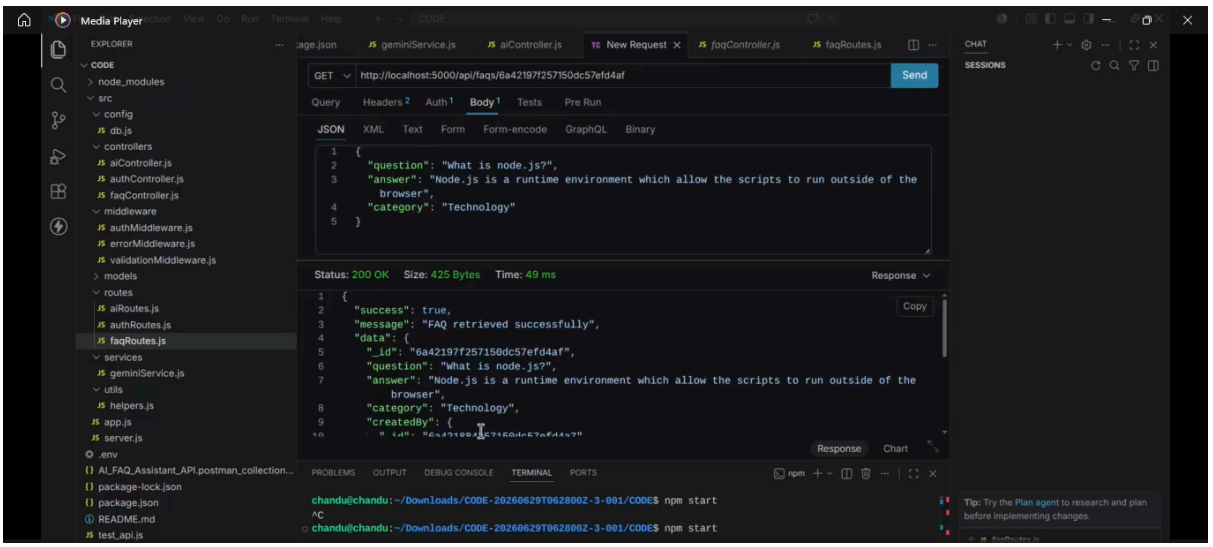
User registration output



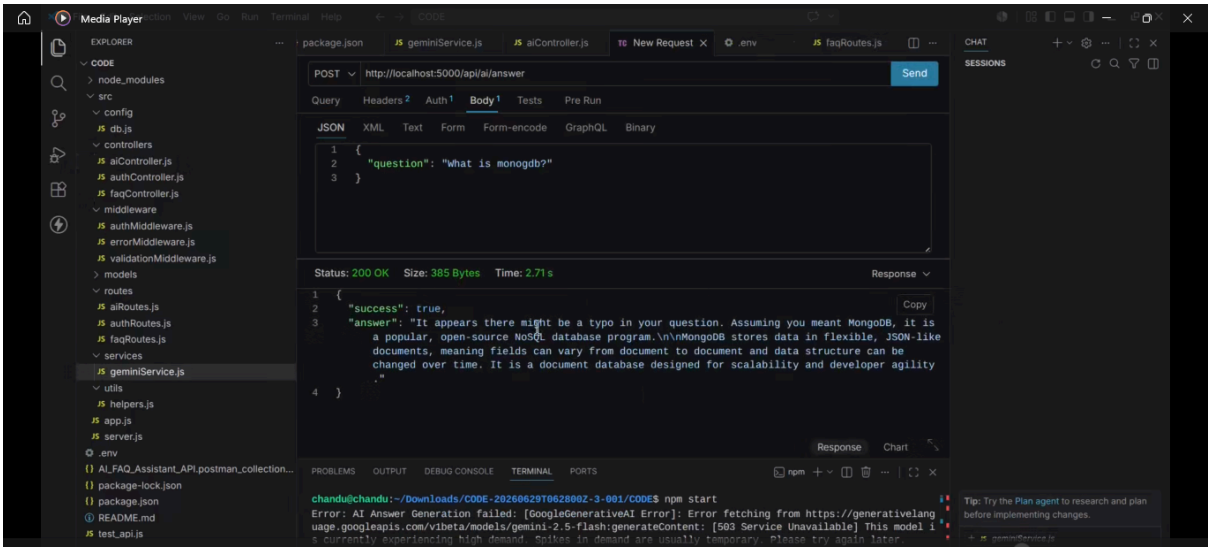
User login output



FAQ Retrieval Output



AI Answer Output



16.ADVANTAGES

The AI FAQ Assistant provides several benefits by combining Artificial Intelligence with backend and database technologies.

Automated Assistance

The system provides automated answers to frequently asked questions, reducing the need for continuous manual support.

Quick Response

Users can submit questions and receive responses quickly through the API-based system.

Easy FAQ Management

Frequently asked questions can be stored and retrieved efficiently using MongoDB.

AI-Based Answering

The integrated AI service can process user questions and generate suitable responses.

Secure User Access

User registration and authentication help control access to the application.

Easy Testing

The REST APIs can be easily tested and verified using Postman.

Scalable Design

The modular architecture allows additional features and services to be added in the future.

17.LIMITATIONS

The AI FAQ Assistant has some limitations that can be improved in future versions.

Internet Dependency

The AI service requires a stable internet connection to process questions and generate responses.

AI Service Availability

The availability of the AI-based response depends on the external AI service. Temporary service errors may affect the response.

Limited Knowledge Source

The system mainly uses the stored FAQ information and connected AI service. It may not provide accurate answers for every type of question.

API-Based Interface

The current project mainly focuses on backend APIs and testing through Postman rather than providing a complete user-friendly frontend interface.

Security Improvements

Additional security features such as advanced encryption, stronger access control, and improved API security can be added in future versions.

18.FUTURE ENHANCEMENT

The AI FAQ Assistant can be further improved by adding advanced features and technologies to make the system more useful and user-friendly.

User-Friendly Frontend

A complete web-based frontend can be developed so that users can interact with the AI FAQ Assistant without using API testing tools.

Voice-Based Interaction

Voice input and voice output can be added to allow users to ask questions and receive answers through speech.

Multilingual Support

Support for multiple languages can be introduced so that users can interact with the system in their preferred language.

Improved AI ResponseS

Advanced AI models and a larger knowledge base can be integrated to improve the accuracy and relevance of generated answers.

Enhanced Security

Additional security features such as stronger authentication, role-based access, and improved data protection can be implemented.

Analytics and Chat History

A feature for storing chat history and displaying frequently asked questions can be added to help analyze user interactions.

Overall, these enhancements can make the

19.CONCLUSION

The AI FAQ Assistant was successfully developed as an intelligent system for providing automated answers to frequently asked questions. The project combines Node.js, Express.js, MongoDB, and an AI service to provide an efficient question-answering solution.

The system supports user registration and authentication, FAQ management, database operations, and AI-based question answering. The APIs were tested using Postman to verify the functionality and responses of the application.

The project demonstrates how Artificial Intelligence and modern backend technologies can be integrated to automate FAQ management and provide quick responses to users. With further enhancements such as a user-friendly frontend, multilingual support, voice interaction, and improved security, the system can be extended for wider real-world applications.